Paper 2: Radius Sweep of Fully-Inscribed Unit-Cell Counts on a 4-Dimensional Lattice

An Enumeration Table from Radius 0.5 to 10.0 and a Reproducible Formulation

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Abstract

In this paper we count the number of 4-dimensional unit cells of side 1, placed on a 4-dimensional integer lattice, that are fully inscribed in a 4-dimensional hyperball of radius R. The object is a purely geometric / integer-lattice counting problem, and we do not treat any correspondence with physical constants.

Taking the center of a unit cell to be

$$k = (k_1, k_2, k_3, k_4) \in \mathbb{Z}^4,$$

with half-width 1/2 in each direction, the condition that this unit cell be fully inscribed in the hyperball of radius R is

$$\sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \le R^2.$$

We define the number of lattice points satisfying this condition as

$$N_0(R) = \# \left\{ k \in \mathbb{Z}^4 \mid \sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \le R^2 \right\}.$$

In this paper we compute $N_0(R)$ for radii $R = 0.5, 1.0, 1.5, \ldots, 10.0$, and tabulate the diameter $2R$ of the circumscribing ball and the diagonal length $2\rho(R)$ of the actually stacked cell set.

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1. Purpose

In Paper 1 we showed that choosing a 4-dimensional lattice in the wavelength space or the frequency space turns the sum-of-squares condition into a unit-cell counting problem.

In this paper we actually carry out that counting.

What we treat here is not a physical theory but the following purely geometric problem.

> Among the unit 4-cells of side 1 on the 4-dimensional integer lattice, how many are fully inscribed in the 4-dimensional hyperball of radius R?

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2. Basic Definitions

Let the 4-dimensional integer lattice be

$$\mathbb{Z}^4.$$

To each lattice point

$$k = (k_1, k_2, k_3, k_4)$$

associate a 4-dimensional unit cell of side 1.

When the cell center is at k, each point x_i inside the cell satisfies

$$k_i - \frac{1}{2} \leq x_i \leq k_i + \frac{1}{2}.$$

The squared distance from the origin to the farthest vertex of this cell is

$$r_{\max}(k)^2 = \sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2.$$

Hence the condition that this cell be fully inscribed in the hyperball of radius R is

$$r_{\max}(k)^2 \leq R^2,$$

that is,

$$\sum_{i=1}^{4} \left(|k_i| + \frac{1}{2}\right)^2 \leq R^2.$$

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3. Fully-Inscribed Cell Count

We define the fully-inscribed cell count as

$$N_0(R) = \# \left\{ k \in \mathbb{Z}^4 \mid \sum_{i=1}^{4} \left(|k_i| + \frac{1}{2}\right)^2 \leq R^2 \right\}.$$

Multiplying both sides by 4,

$$\sum_{i=1}^{4} (2|k_i| + 1)^2 \leq (2R)^2.$$

Hence this problem can also be expressed as an integer inequality on a sum of four positive odd squares.

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4. Stacked Diagonal Length

Let the set of cells inscribed at radius R be

$$K_R = \left\{ k \in \mathbb{Z}^4 \mid r_{\max}(k) \leq R \right\}.$$

Among the cells in this set, define the distance from the origin to the farthest vertex as

$$\rho(R) = \max_{k \in K_R} r_{\max}(k).$$

If K_R is empty, we set

$$\rho(R) = 0.$$

The centrally symmetric maximal diagonal length of the stacked cell set is then defined as

$$L(R) = 2\rho(R),$$

and the diameter of the circumscribing ball is

$$D(R) = 2R.$$

Hence always

$$L(R) \leq D(R).$$

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5. Radius Sweep Results

Table 1 shows the results computed from radius $R = 0.5$ to 10.0 in steps of 0.5.

Table 1. Radius sweep of fully-inscribed unit-cell counts on the 4-dimensional lattice

Circumscribing radius R	Diameter 2R	Fully-inscribed cell count N0(R)	Stacked diagonal $2\rho(R)$	Radial margin $R - \rho(R)$
0.5	1	0	0	0.5
1	2	1	2	0
1.5	3	1	2	0.5
2	4	9	3.4641	0.267949
2.5	5	33	4.47214	0.263932
3	6	137	6	0
3.5	7	233	6.63325	0.183375
4	8	473	7.74597	0.127017
4.5	9	809	8.7178	0.141101
5	10	1545	10	0
5.5	11	2233	10.7703	0.114835
6	12	3457	11.8322	0.0839202
6.5	13	5001	12.8062	0.0968758
7	14	7281	14	0
7.5	15	9489	14.8324	0.0838015
8	16	12833	15.8745	0.0627461
8.5	17	16657	16.8523	0.0738502
9	18	22409	18	0
9.5	19	27233	18.868	0.0660189
10	20	34569	19.8997	0.0501256

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6. Basic Check Values

Extracting from Table 1 the first values at integer radius,

$$N_0(1) = 1,$$

$$N_0(2) = 9,$$

$$N_0(3) = 137,$$

$$N_0(4) = 473,$$

$$N_0(5) = 1545.$$

In particular,

$$N_0(3) = 137$$

is obtained as the number of unit 4-cells of side 1 fully inscribed in the 4-dimensional hyperball of radius 3.

This 137 is not introduced under any assumed correspondence with a physical constant; it is a pure counting result obtained from the 4-dimensional integer lattice, radius 3, the unit cell, and the full-inscription condition.

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7. Decomposition at $R = 3$

For $R = 3$, the condition is

$$\sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \leq 9.$$

Multiplying both sides by 4,

$$\sum_{i=1}^{4} (2|k_i| + 1)^2 \leq 36.$$

The possible shells of sums of odd squares in this range are

$$4,\ 12,\ 20,\ 28,\ 36.$$

The counts per shell are

$$4:1, \quad 12:8, \quad 20:24, \quad 28:40, \quad 36:64.$$

Hence

$$1+8+24+40+64=137.$$

Note that the shell 28 is the combined count of the $(3,3,3,1)$ type and the $(5,1,1,1)$ type.

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8. Reproduction Algorithm

Table 1 can be reproduced by the following pseudocode.

```
def count_cells(R):
    count = 0
    max_s = 0
    max_abs = floor(R - 0.5) + 1

    for k1 in range(-max_abs, max_abs + 1):
      for k2 in range(-max_abs, max_abs + 1):
        for k3 in range(-max_abs, max_abs + 1):
          for k4 in range(-max_abs, max_abs + 1):
            s = sum((abs(ki) + 0.5)**2 for ki in [k1,k2,k3,k4])
            if s <= R**2:
                count += 1
                max_s = max(max_s, s)

    rho = sqrt(max_s) if count > 0 else 0
    diagonal = 2 * rho
    return count, rho, diagonal
```

This computation merely judges, for each cell, whether its farthest vertex lies within the ball of radius R.

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9. On the Absence of Uncertainty

This paper does not treat the weighted contribution of near-boundary cells or the effective counting including the uncertainty width δ.

That is, what is computed in this paper is the fully-inscribed counting at

$$\delta = 0.$$

To perform counting with uncertainty, the indicator function must be replaced by a weight function

$$W_\delta(x).$$

However, how to determine the value of δ or the form of that function is undefined and outside the scope of this paper.

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10. Conclusion

In this paper we computed, from radius 0.5 to 10.0 in steps of 0.5, the number of unit 4-cells of side 1 on the 4-dimensional integer lattice that are fully inscribed in the 4-dimensional hyperball of radius R.

The fully-inscribed cell count is defined by

$$N_0(R) = \#\left\{ k \in \mathbb{Z}^4 \mid \sum_{i=1}^{4} \left(|k_i| + \frac{1}{2} \right)^2 \leq R^2 \right\}.$$

This counting is purely geometric / integer-lattice in nature and asserts no correspondence with physical constants.

In particular,

$$N_0(1) = 1, \qquad N_0(2) = 9, \qquad N_0(3) = 137$$

are obtained.

The results of this paper provide a basic reference showing that the sum-of-squares constraint, when the wavelength space or the frequency space is chosen as a 4-dimensional lattice, can be treated as a concrete unit-cell counting problem.

Appendix A: CSV File

Table 1 of this paper is also saved as the accompanying file

`paper2_radius_sweep_R_0_5_to_10_step_0_5.csv`.

The meaning of the columns is as follows.

- `R`: radius of the circumscribing ball
- `sphere_diameter_2R`: diameter $2R$ of the circumscribing ball
- `full_inclusion_cell_count_N0`: fully-inscribed cell count $N_0(R)$
- `stack_vertex_radius`: maximal vertex radius $\rho(R)$ of the stacked cell set
- `stack_diagonal_length`: stacked diagonal length $2\rho(R)$
- `unused_margin_R_minus_stack_radius`: radial margin $R - \rho(R)$
- `shells_S_counts`: count per shell $S = \sum_i (2|k_i| + 1)^2$